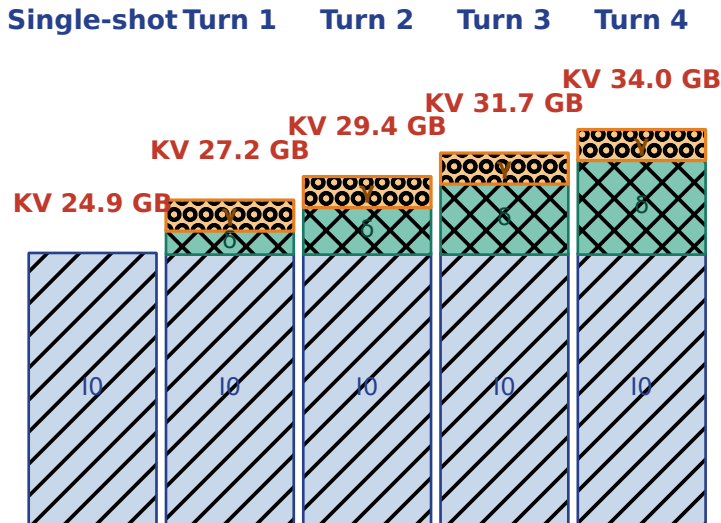


Agentic context accumulation: the KV block grows every turn



append-only illustrative model: context is only appended, never evicted

agentic depth is a fleet-sizing problem: KV grows ~36% (24.9 → 34.0 GB)

canonical example @ 2.62 MB/token: total = $10 + T \cdot \delta + T \cdot \gamma + O_{\text{final}}$ (base 9.5K = 9.2K input + 300 output).

 initial prompt (10)  retrieved tool output (δ)  generated reasoning (γ)